

Dataset Documentation

Find Evil! — Autonomous DFIR Triage Agent

This document records exactly what the agent was tested against, where the data comes from, how to obtain and verify it, and what the agent found. It is the starting point for reproducing our results.

All test data are public, standard forensic teaching images. We use them read-only: each image is set chmod 444, a SHA-256 baseline is recorded before analysis and re-verified afterward, and the agent is permitted only read-only Sleuth Kit operations.

OVERVIEW

Datasets at a Glance

Dataset	Source	Format	Investigation type	Outcome
CFReDS Hacking Case (Greg Schardt / “Mr. Evil”)	NIST CFReDS	EnCase .E01/.E02 and raw SCHARDT.001...008	Intrusion / attacker tooling	Attacker staging system
M57-Jean (nps-2008-jean)	Digital Corpora	EnCase .E01/.E02 (≈3 GB)	Data exfiltration / social engineering	Social engineering → data exfiltration

Both cases are solved by the same single generalist agent; it determines the investigation type from the evidence rather than being told which case it is looking at.

DATASET

Dataset 1 — CFReDS Hacking Case (“Mr. Evil”)

Source

NIST Computer Forensic Reference Data Sets (CFReDS) project — the “Hacking Case.” Publicly distributed by NIST for forensic training and tool testing.

- Project: <https://cfreds.nist.gov/>
- Case landing (legacy): <https://cfreds.nist.gov/all/NIST/HackingCase>

Scenario (as published by NIST)

On 09/20/04 a Dell CPI notebook (serial # VLQLW) was found abandoned together with a wireless PCMCIA card and a homemade 802.11b antenna. The machine is suspected of being used for wireless “war-driving” — parking within range of Wi-Fi hotspots (e.g. T-Mobile/Starbucks) to intercept traffic and harvest credit-card numbers, usernames and passwords. The suspect, Greg Schardt, uses the online handle “Mr. Evil.” The stated

objectives are to find hacking software, evidence of its use, and any generated data, and to tie the machine to the suspect.

Image files

- EnCase / Expert Witness: Dell Latitude CPi.E01, Dell Latitude CPi.E02
- Raw dd split: SCHARDT.001 ... SCHARDT.008 (plus a notes file)
- The EnCase set and the raw set are the same physical disk — the agent produces equivalent results from either. We use the .E01 as the primary demo image.

Observed acquisition facts (from our run)

- Partition table via mmls: NTFS partition at sector offset 63.
- Filesystem: NTFS, Windows host; primary user account “Mr. Evil.”
- NTFS volume serial: B26CB1CE6CB18D9B (identical across both image formats, confirming they are one disk).
- NIST publishes official acquisition MD5/SHA hashes on the case page; our pipeline additionally records its own SHA-256 baseline at run time for chain-of-custody.

What the agent found

Classified the laptop as an attacker staging system (Intrusion track). Key findings, each cited to a filesystem inode and source artifact:

- **Cain & Abel — CONFIRMED EXECUTED** (prefetch artifact CAIN.EXE-23D61279.pf), not merely present.
- Credential theft: pwdump, pwdump2, SAMDUMP.
- Network reconnaissance: nmap / nmapNT, nbtscan, enum.
- Anti-forensics: ELSAVE (Windows event-log wiper).
- Privilege escalation: GetAdmin, SECHOLE.
- Remote access: NetBus RAT, WinVNC / vncviewer.
- Other tooling: netcat (nc.exe), John the Ripper, Brutus.
- Deleted, recovered from the Recycle Bin (INFO2): wireless / packet-capture installers — NetStumbler, Ethereal, WinPcap — consistent with prior war-driving activity.
- Attribution: browser / webmail artifacts tying activity to “Mr. Evil” Yahoo / Hotmail accounts.

This aligns with the published CFReDS answer key: a fully assembled wireless-interception and credential-theft toolkit owned by Greg Schardt / “Mr. Evil.”

DATASET

Dataset 2 — M57-Jean (nps-2008-jean)

Source

Digital Corpora — the “2009 M57-Jean” scenario, a component of the larger M57-Patents corpus. Freely available for research and education.

- Scenario page: <https://digitalcorpora.org/corpora/scenarios/m57-jean/>

Scenario (as published)

A small start-up, M57.biz, discovers that a confidential spreadsheet containing the names and salaries of key employees has been posted to the “comments” section of a competitor's website. The spreadsheet existed on only one machine — that of an officer named Jean — who claims she has no idea how the data left her laptop and believes she was hacked. The task is to determine how the data was exfiltrated, and whether Jean is as innocent as she claims.

Image files

- EnCase / Expert Witness: nps-2008-jean.E01, nps-2008-jean.E02 (multi-volume EWF, ≈3 GB total).
- Both segments are required; they are one logical image.

What the agent found

Classified the incident as social engineering → data exfiltration (Exfiltration track) — explicitly not an intrusion or malware case, since no attacker toolkit is present. Findings:

- Sensitive document: the spreadsheet m57biz.xls (employee names, salaries, SSNs).
- Social-engineering pretext: a request that appeared to come from alison@m57.biz (a company officer), while the actual external correspondent was tuckgorge@gmail.com.
- Outbound disclosure: Jean emailed m57biz.xls to the external Gmail address.
- Generic detection logic: the collector auto-detected the organisation's home domain (m57.biz) as the most frequent email domain on disk, then flagged any message carrying an attachment to an address outside that domain — nothing about this case is hard-coded.

This matches the established M57-Jean ground truth: Jean was socially engineered (spear-phished) into emailing the spreadsheet to an outsider; the exfiltration channel was email.

SCOPE

Data Deliberately Out of Scope

The following items sometimes appear alongside these cases but are not disk images and are excluded from the agent's disk-triage scope by design:

- nitroba.pcap — a network capture, not a filesystem (would require a packet-analysis track).
- Dc1.exe and similar standalone binaries — single executables, not mountable images.

They are noted here so reviewers know the boundary of what was tested.

HOW TO REPRODUCE

Reproducibility & Evidence Integrity

1. Obtain the images from the sources above (NIST CFReDS; Digital Corpora). Both are free.

Lock them read-only and baseline them before any analysis:

```
chmod 444 <image>
sha256sum <image> > <image>.sha256          # baseline (chain of custody)
```

Phase 1 — deterministic collection (no LLM):

```
bash collect.sh "<image>" bundle/
```

collect.sh auto-detects the partition offset from mmls (offset 63 on these NTFS volumes) and builds a text “evidence bundle” using read-only Sleuth Kit tools.

Phase 2 — autonomous reasoning over the bundle via Claude Code (claude -p, --permission-mode dontAsk), governed by CLAUDE.md.

Verify integrity after the run:

```
sha256sum -c <image>.sha256 # must report OK – image unchanged
```

Because the images are chmod 444 and write-class commands are denied by the agent's permission configuration, the post-run hash matches the baseline on every run (no spoliation).

Format note

For the Schardt disk, the EnCase .E01 and the raw SCHARDT.001... split are the same physical disk (identical NTFS volume serial), so either format reproduces the same findings; we recommend the .E01 as the primary image.

REFERENCES

Sources

- NIST CFReDS project — <https://cfreds.nist.gov/>
- NIST CFReDS Hacking Case — <https://cfreds.nist.gov/all/NIST/HackingCase>
- Digital Corpora, 2009 M57-Jean scenario — <https://digitalcorpora.org/corpora/scenarios/m57-jean/>